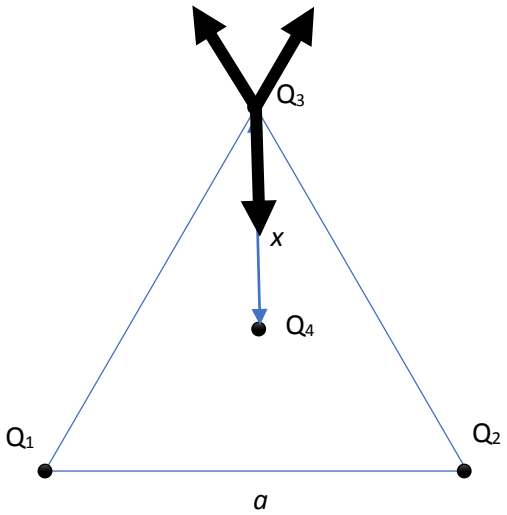
**Question 1**

a)




b) $F_{TOT} = F_4 + 2F_1(vertical\ components)$ or comment on symmetry

$$= \frac{k(-q)q}{x^2} + 2\frac{k(-q)(-2q)}{a^2}\frac{\sqrt{3}}{2}$$

$$= -\frac{kq^2}{x^2} + 2\sqrt{3}\frac{kq^2}{a^2}$$

$$= kq^2\left(\frac{2\sqrt{3}}{a^2} - \frac{1}{x^2}\right)$$

c) $x = \frac{1}{2}a, x^2 = \frac{1}{4}a^2$

$$F_{net} = kq^2\left(\frac{2\sqrt{3}}{a^2} - \frac{4}{a^2}\right)$$

Therefore direction is downwards or towards $Q_4$

d) $\frac{2\sqrt{3}}{a^2} = \frac{1}{x^2}$

$$x^2 = \frac{a^2}{2\sqrt{3}}$$

$$x = \frac{a}{\sqrt{2}\sqrt[4]{3}}$$

**Question 2**

a) i) $E = (2.0x^2\hat{\imath} - 3.0\hat{\jmath})$

$$\emptyset_{TOP} = \int E \cdot dA_y$$

$$= \int (2.0x^2\hat{\imath}\cdot\hat{\jmath} - 3.0\hat{\jmath}\cdot\hat{\jmath})dA_y$$

$$= 0 - 3.0\int dA_y$$

$$= -3.0 \times 4.0$$

$$= -12\ Nm^2C^{-1}$$

ii) $\emptyset_{RIGHT} = \int E \cdot dA_x$

$$= \int (2.0x^2\hat{\imath} \cdot \hat{\imath} - 3.0\hat{\jmath} \cdot \hat{\imath}) dA_x$$

$$= \int (2.0x^2 - 0)\, dA_x, x = 2$$

$$= 2.0 \times 4.0 \int dA_x$$

$$= 8 \times 4 = 32\ Nm^2C^{-1}$$

b) bottom $\phi$ = 12 $Nm^2C^{-1}$

$E_y$ is same, $dA_y$ is in -ve j direction

left $\phi$ = 0 $Nm^2C^{-1}$

x = 0 => $E_x$ = 0

front $\phi$ = 0 $Nm^2C^{-1}$

as $E_z$ = 0

back $\phi$ = 0 $Nm^2C^{-1}$ as above

Total $\phi$ = -12 + 12 + 32 + 0 + 0 + 0 = 32 $Nm^2C^{-1}$

c) $\emptyset_{tot} = \frac{q_{enclosed}}{\varepsilon_0}$

$$q_{enclosed} = 8.85 \times 10^{-12} \times 32 = 2.8 \times 10^{-10}\ C$$

d)$V_{B-A} = -\int_A^B E \cdot ds$

$$= -\int_0^2 E \cdot dx\hat{\imath}$$

$$= -\int_0^2 (2.0x^2\hat{\imath} \cdot \hat{\imath} - 3.0\hat{\jmath} \cdot \hat{\imath}) dx$$

$$= -\int_0^2 2.0x^2 dx$$

$$= -\left[\frac{2}{3}x^3\right]_0^2 = -\frac{16}{3} = -5.3V$$

e) Yes
E-field is non-uniform not spherical distribution around (1,1,1)
Therefor changing shape of the area will alter the electric flux through the surface

**Question 3**

a) Current Conservation: 0.05 + $I_1$ = $I_2$

Loop 1: clockwise 3 – $I_1$x10 – 6 – $I_2$R - $I_1$x10 = 0

$20I_1 + I_2R = -3$

Loop 2: anticlockwise $6 – 0.05x10 – 6 - I_2R – 0.05x10 = 0$

$-1 = I_2R$

Hence I1 = -0.10 A

I2 = -0.05 A

R = 20 Ω

b) $V_{AB}$ = - 6 + 20x0.05 = -5 V (middle wire)
$V_{AB}$ = -3 – 10x0.1 – 10x0.1 = -5 V (left wire)
$V_{AB}$ = -6 + 10x0.05 + 10x.05 = 5 V (right wire)

**Question 4**

For this question, key information is listed below. Allocation of information to sections as listed is not rigid – the inclusion of the information will still warrant marks.

Faraday Cage

i. When you place a radio-wave transmitter (or receiver) in an enclosed metal cage the signal does not leave or go into the metal cage (appropriate diagram helpful). Also descriptions of a car or plane getting hit by lightning.
ii. $\oint \boldsymbol{E}.d\boldsymbol{A} = \frac{q}{\varepsilon_o}$
iii. Includes the following points;
    a. A conductor has zero electric field inside due to any electric field inside a conductor will cause the free electrons to redistribute themselves to create a zero electric field again – Electrostatics
    b. An electric field incident on a metal cage will cause the electrons to move such that they counter the incident Electric field resulting in a zero field inside.
    c. Holes in metal need to be smaller than wavelength – hence we can usually see through the cage
    d. Any diagrams to help with the explanation.
iv. EMP protection, microwaves, cars and aeroplanes (other possible examples are ok, too)

Meisner Effect

i. When you place a magnet above a superconductor at law temperatures the magnet will levitate (include appropriate diagram).
ii. $\oint \boldsymbol{B}.d\boldsymbol{A} = 0$
iii. Include the following points;
    a. A superconductor has zero resistance, hence the electrons move freely in the superconductor. Once moving, the electrons can maintain their current for a very long time.
    b. Introducing a magnet near the superconductor sets up eddy currents in the superconductor
    c. In an external magnetic field a cooled superconductor will have no B field inside and an opposing B field will be created on the skin of the superconductor repelling the external field.
    (diagrams would be helpful for all these explanations.)

d. Mentioning perfect diamagnetism and or Cooper pairs.

iv. Levitating magnets, high speed bullet trains [other possible examples are ok, too]

Displacement Current

i. We detect a magnetic field[1] between the plates of a charging capacitor

ii. $\oint \boldsymbol{B}.d\boldsymbol{s} = \mu_o i + \mu_o i_D$

iii. They should discuss the following points;

a. Exists when you are charging a capacitor such that an imaginary current occurs between the capacitor plates despite there being no movement of charge.

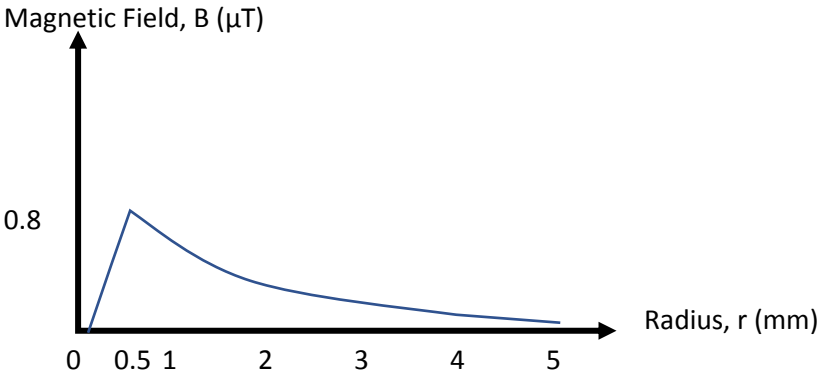
b. Confirmed due to the detection of the magnetic field between the plates only while the capacitor is charging.

c. Using Maxwell's proposal of symmetry and that a changing E field must create a B field, hence; $i_D = \varepsilon_o \frac{d\Phi_E}{dt}$
(Diagrams help with the explanations.)

iv. Allows an open capacitor to behave as a wire while charging.

**Question 5**

a)




Requires appropriate axis labels and scale, and appropriate graph shape.

Use of Ampere's law to determine B at r=0.5 mm, B = 0.8 μT

B=0 for r<0.1 mm

b) Straight sections of wire parallel to ds – B=0

Curved section perpendicular to ds

$$db = \frac{\mu_0 i ds \sin(0)}{4\pi R^2}$$

$$B = \int dB = \frac{\mu_0 i}{4\pi R^2} \int_0^{\pi R} ds$$

$$= \frac{\mu_0 i}{4\pi} = 1.3 \times 10^{-7} T = 0.13 \mu T$$

c) $B = \mu_0 ni \Rightarrow n = \frac{B}{\mu_0 i} = 198{,}940$ turns per metre

d) $\mu = \frac{B}{ni} = 6.3 \times 10^{-3} T\ m\ A^{-1}$

e) The following points should be included

   a. Faraday's Law does stipulate that an emf will be induced only due to a changing magnetic flux in a wire loop. A changing emf or current will also induce a changing magnetic field.
   b. But a current in a coil, such as the solenoid can create a magnetic field as shown in parts a, b) even if the current is NOT changing – the resulting magnetic field is constant/steady.
   c. Hence with a direct current passing through the coil, it will create a steady magnetic field, boosted by the iron core which can then attract the lever and close the switch.
   d. As long as the coil has current, the switch will be closed.
   e. Ivory is correct

**Question 6**

a) According to the hydrostatic equation:
   $P_A$=2 atm
   $P_1 = P_A + \rho g\ z_A$ where $z_A$ is the height of point A from point 1. We set z=0 for 1 and 2.

b) $P_2 = P_B + \rho g z_B = P_B + \rho g(z_A + \Delta z)$ where $P_B$ = 1 atm and $\Delta z$ = 0.5 m

c) $\Delta P = P_1 - P_2 = P_A - P_B - \rho g \Delta z = \left(2 - 1 - \frac{1000 \times 9.8 \times 0.5}{101325}\right) atm = 0.95\ atm$
   The system is not at equilibrium – a jet of water traverses the duct from 1 to 2.
   In 2, the streamlines are parallel to each other and horizontal to the ground, so the hydrostatic formula in Part (b) holds.
   From Bernoulli's equation, applied to a streamline connecting a point of the surface "A" with a point at "2", we have:

$$P_A + \frac{1}{2}\rho v_A^2 + \rho g\ z_A = P_2 + \frac{1}{2}\rho v_2^2$$

   Since the fluid is incompressible,

$$(Xsection\ Area\ of\ A) \times\ v_A = (Xsection\ Area\ of\ 2) \times v_2$$

   As Area(A)>>Area(2), so $v_2 >> v_A$.
   The term $\frac{1}{2}\rho v_A^2$ is therefore negligible (≈0), reducing our relation to:

$$\frac{1}{2}\rho v_2^2 = P_A - P_2 - \rho g z_A = P_A - P_B - \rho g \Delta z$$

$$\therefore v_2 = \left(\frac{2}{\rho} \times 0.95 \times 101325\right)^{\frac{1}{2}} = 13.9\ ms^{-1}$$

   Since the cross-section of the duct is constant (and at constant height), the velocity is constant along the duct.

d) It is not, as long as this section at point 2 remains constant. The velocity would be proportional to the inverse of the section along the duct.

e) As $v_1$=$v_2$ and $z_1$ = $z_2$ = 0, so from the Bernoulli equation, we have $P_1$=$P_2$.

**Question 7**

**a) i)** PV = nRT

$$n = \frac{PV}{RT} = \frac{101325 \times 0.1}{8.31 \times 288} = 4.2337 \text{ mole (before)}$$

OR, $n = \frac{PV}{RT} = \frac{101325 \times 0.11}{8.31 \times 316} = 4.2445 \text{ mole (after)}$

(Not able to actually find Molar Mass as requested in original exam)

ii) Isobaric → pressure a constant so $W = -\int PdV \rightarrow W = -P\Delta V$

$$W = -P(\mathrm{V}_2 - V_1) = -1.013 \times 10^3 \text{ J} = -1 \text{ kJ}$$

iii) $\Delta E = Q + W = (3.7 - 1.1) \times 10^3 \text{ J} = 2.7 \text{ kJ}$

iv) $Q = nC_P\Delta T \rightarrow C_p = \frac{Q}{n\Delta T}$

As a monatomic gas has $C_P$=(5/2)R and a diatomic gas has $C_P$=(7/2)R,

$\frac{C_P}{R} = \frac{Q}{Rn\Delta T} = 3.76 \sim \frac{7}{2} \rightarrow likely\ \boldsymbol{DIATOMIC}$

v) Yes, if Q=-W (such as in an isothermal process), then ΔE = 0 → ΔT=0

Similarly, an increase in volume is work done by the gas (-W) which could be offset by addition of energy by heating +Q.

b) Equipartition of gases:

- Describes how energy is stored by gases due to the "degrees of freedom"
- Each degree of freedom is independent, as is the corresponding stored energy – (1/2)$k_B$T per molecule.
- Degrees of freedom are associated with movement of atom/molecule – whether 3 (the translational degrees along x,y and z) or also rotational and vibrational (extra 2 degrees).

**Question 8**

a) $\lambda_{max} = 293 \times 10^{-9} m$

$E_{min} = hf_{min} = \frac{hc}{\lambda_{max}} = 4.24\ eV$

$\rightarrow unknown\ metal = Chromium$

$when\ \lambda_{max} \rightarrow KE_{min} = 0 \rightarrow hf_{min} = Work\ function$

And $W_{Ch}$ = 4.24 eV = $hf_{min}$

b) X-Rays $E = 500\ keV;\ \theta = 45°$

i) $\Delta\lambda = \frac{h}{mc}(1 - \cos\theta) = 7.11 \times 10^{-13}\text{m} = 0.71\text{pm}$

ii) $E = 500\ keV = \frac{hc}{\lambda} \rightarrow \lambda = 2.484 \times 10^{-12}\text{m}$

$\rightarrow if\ e\ is\ recoiling, E_\chi \downarrow \therefore \lambda_\chi \uparrow, \quad \therefore\ \lambda + \Delta\lambda = 3.195 \times 10^{-12} m$

$$E'_\chi = \frac{hc}{3.195 \times 10^{-12}} = 389 \text{ keV}$$

ALSO expressed as follows:

$$E_{\chi_2} = \frac{E_{\chi_1}}{1 + \left(\frac{E_{\chi_1}}{m_e c^2}\right)(1 - \cos\theta)} = 389 \text{ keV}$$

iii) If solely kinetic energy of the electron: $E(electron) = \Delta E_\chi = 111.2\ \text{keV}$
If including the rest mass energy (relativistic): $E = \Delta E_\chi + m_e c^2 = 622\ \text{keV}$

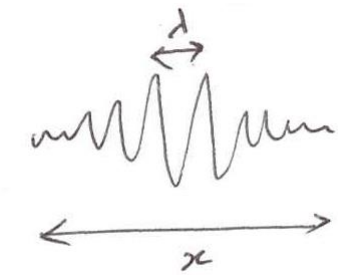
iv) Non-relativistic method:

$$v_e = \sqrt{\frac{2E}{m}} = 1.98 \times 10^8 ms^{-1} \rightarrow \lambda = \frac{h}{mv} = 3.68 \times 10^{-12} m$$

$$\frac{\lambda_e}{\lambda_\chi} = 1.15$$

→ same order of magnitude – this is the case in which Compton scattering is significant.
If $E_{\chi_1} >> m_e c^2, E_{\chi_2} \sim 0$ at any angle ≠ 0 (no scattering)
If $E_{\chi_1} << m_e c^2, E_{\chi_2} \sim E_\chi_1$ and the electron has no recoil.

c) Heisenberg Uncertainty Principle $\Delta x \Delta p \geq \frac{\hbar}{2}$




If we find x within a narrow value, i.e. Δx is small, the wavelength $\lambda$ will be **small**.
But, as momentum $p = \frac{h}{\lambda}$, the uncertainty with momentum $\Delta p$ will be **large**.

d) $\frac{\Delta\lambda}{\lambda} = \frac{\Delta p}{p} = 10^{-7} \rightarrow p = \frac{h}{\lambda} = 1.33 \times 10^{-26}\ \text{kgms}^{-1}$

$$\therefore \Delta p = p \times 10^{-7} = 1.33 \times 10^{-33}\ \text{kgms}^{-1}$$

$$\rightarrow \Delta x = \frac{\hbar}{2}\frac{1}{\Delta p} = 0.0398\ \text{m}$$

**Question 9**

a) T=3000K

i. $\lambda_{max} T = 2898 \mu m\ K \rightarrow \lambda_{max} = 966\ nm$

ii. Surface area A $= 4 \times 10^{21}\ km^2 = 4 \times 10^{27}\ m^2$
→ Stefan relation $P = \sigma \varepsilon A T^4$ where $\varepsilon = 1.0, \sigma = 5.67 \times 10^{-8}\ \text{Wm}^{-2}\text{K}^{-4}$
$P = 1.84 \times 10^{34}\text{W}$

iii. $S_\lambda(T) = \frac{2c^2 h}{\lambda^5}\left(\frac{1}{\exp\left(\frac{hc}{\lambda k_B T}\right) - 1}\right)$

$$= 1.4187 \times 10^{14} \times \left(\frac{1}{144.53 - 1}\right)$$

$$= 9.885 \times 10^{11}\ W\ /A\ /\lambda$$

[Note the question asks for Spectral POWER, not radiancy
So → $P_{\lambda_{max}} = S_\lambda A = 3.95 \times 10^{39} W\ /\ \lambda$ ]

b) i) $\lambda = 966nm \rightarrow E_\gamma = \frac{hc}{\lambda} = 2.057 \times 10^{-19} J = 1.29 eV$

ii) $\#photons/\sec = \frac{P}{E_\gamma} = \frac{1.84 \times 10^{34}}{2.057 \times 10^{-19}} = 8.93 \times 10^{52}\ photons/second$

(Also accepted in exam: $\frac{\#\gamma}{s} = \frac{P_{\lambda_{max}} \times \lambda_{\max}}{E_\gamma} = \frac{3.816 \times 10^{33}}{2.057 \times 10^{-19}} = 1.85 \times 10^{52}\ \gamma/s$ )

**Question 10**

a) Quantised angular momentum – discrete orbital position
Classical view – if all orbits allowed, as the electrons spin around nucleus, accelerate due to circular motion → EM (Maxwell) says this causes energy loss ->released as photons, as a continuous spectra of wavelengths,
→ classical orbits are not stable, as energy lost, electrons would spiral into nucleus.
No classical reason for e to orbit in fixed, narrow energy bands, and only emit discrete energy photons. (in contradiction to classical EM prediction above.)

b) i) Bohr radius of muonic atom $a = \frac{h^2\varepsilon_0}{\pi m e^2} = \frac{1}{200}a_0 = 256.6\text{ fm}$

ii) $E = -\frac{me^4}{8\varepsilon_0^2 h^2}\frac{1}{n^2} \therefore E_\mu = 200E_H = \frac{2720\text{eV}}{n^2}$

iii) $E_{n=2\to1} = 200\left(-\frac{13.6}{1} - \left(-\frac{13.6}{4}\right)\right) = 2040\text{ eV}$

iv) The muonic atom is 200 times smaller than the normal electronic hydrogen atom, but the transitions between energy levels are 200 times bigger. (muon is more tightly bound, takes more energy to change to higher orbital levels.)

**Question 11**

Radioactive decays → $R = R_0 e^{-\lambda t}$ where $\lambda = \frac{\log_e 2}{\tau_{\frac{1}{2}}}$ and $\tau_{\frac{1}{2}}$ is the half life of the material.

a) $R(^{14}\text{C}) = (0.23)e^{-1.209\times10^{-4}y^{-1}\times5\,y} = 0.23 \times 0.99939 = 0.229\,Bq$

$R(^{15}\text{O}) = (7\times10^{10})e^{-0.00567\,s^{-1}\times1.578\times10^{8}\,s} = 0\,Bq$

$R(^{3}\text{H}) = (3.6\times10^{14})e^{-0.0562\,y^{-1}\times5\,y} = 2.72\times10^{14}\,Bq$

b) Dating at 5 years – greatest accuracy with these three would be Hydrogen, as there is too little change to be easily measurable in Carbon (<1 part in 100), or two much in Oxygen (~all decayed).

c) As $r_0 = (1.2\times10^{-15})A^{\frac{1}{3}}$, $r_0$(Oxygen) = 2.96 fm
$r_0$(Carbon) = 2.89 fm

d) ${}^{14}_{6}C \to {}^{14}_{7}N + {}^{0}_{-1}\beta + {}^{0}_{0}\bar{\nu}$

$\{Provided\ data\ presents\ some\ difficulties:$
$\Delta E = (14.003241u) - (14.003074u + 0.00055u + 0u) = -3.83\times10^{-4}u.c^2\}$
$\Delta E = \left[(m_n) - \left(m_p + m_e + 0\right)\right]c^2$
$\Delta E = [(1.00866u) - (1.00728u + 0.00055u + 0u)]c^2 = 8.3\times10^{-4}u.c^2$
$= 1.24\times10^{-13}\text{J} = 774\text{ keV}$

e) B.E.$= \left(m_p + 2m_n - m_{\,^3H}\right)c^2 = \Delta mc^2 = 0.0085508u.c^2 = 1.28\times10^{-12}\text{ J} = 8\text{ MeV}$